Name: _____ **Class:** _____

JURONG PIONEER JUNIOR COLLEGE

JC2 Preliminary Examination 2026

PHYSICS **9478/02**
Higher 2
31 August 2026

Paper 2 Structured Questions **2 hours**

You must answer on the question paper.
No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name and class at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- You may use an approved calculator.

INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [ ].

For Examiner's Use	
1	/ 6
2	/ 10
3	/ 9
4	/ 11
5	/ 10
6	/ 10
7	/ 19
Total	/ 75

This document consists of **24** pages. Any blank pages are indicated

Data

speed of light in free space	$c = 3.00 \times 10^{8}$ m s^{-1}
permeability of free space	$\mu_0 = 4\pi \times 10^{-7}$ H m^{-1}
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}$ F m^{-1} $(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}$ m F$^{-1})$
elementary charge	$e = 1.60 \times 10^{-19}$ C
Planck constant	$h = 6.63 \times 10^{-34}$ J s
unified atomic mass constant	$u = 1.66 \times 10^{-27}$ kg
rest mass of electron	$m_e = 9.11 \times 10^{-31}$ kg
rest mass of proton	$m_p = 1.67 \times 10^{-27}$ kg
molar gas constant	$R = 8.31$ J K^{-1} mol^{-1}
Avogadro constant	$N_A = 6.02 \times 10^{23}$ mol^{-1}
Boltzmann constant	$k = 1.38 \times 10^{-23}$ J K^{-1}
gravitational constant	$G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}
acceleration of free fall	$g = 9.81$ m s^{-2}

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
work done on/by a gas	$W = p\Delta V$
pressure	$p = \frac{F}{A}$
gravitational potential	$\phi = -\frac{GM}{r}$
temperature	$T / \text{K} = T / {}^{\circ}\text{C} + 273.15$

pressure of an ideal gas	$p = \frac{1}{3}\frac{Nm}{V}\langle c^2 \rangle$
mean translational kinetic energy of an ideal gas molecule	$E = \frac{3}{2}kT$
displacement of particle in s.h.m.	$x = x_0 \sin \omega t$
velocity of particle in s.h.m.	$v = v_0 \cos \omega t = \pm\omega\sqrt{x_0^2 - x^2}$
electric current	$I = nAvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$
capacitors in series	$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \ldots$
capacitors in parallel	$C = C_1 + C_2 + \ldots$
energy in a capacitor	$U = \frac{1}{2}QV = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}CV^2$
charging a capacitor	$Q = Q_0\left[1 - e^{-\frac{t}{\tau}}\right]$
discharging a capacitor	$Q = Q_0\, e^{-\frac{t}{\tau}}$
RC time constant	$\tau = RC$
electric potential	$V = \frac{Q}{4\pi\varepsilon_0 r}$
alternating current/voltage	$x = x_0 \sin \omega t$
magnetic flux density due to a long straight wire	$B = \frac{\mu_0 I}{2\pi d}$
magnetic flux density due to a flat circular coil	$B = \frac{\mu_0 NI}{2r}$
magnetic flux density due to a long solenoid	$B = \mu_0 nI$

energy states for quantum particle in a box $E_n = \frac{h^2}{8mL^2} n^2$

radioactive decay $x = x_0 e^{-\lambda t}$

radioactive decay constant $\lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$

Answer **all** questions in the spaces provided.

1 An object of surface area A at thermodynamic temperature T will emit electromagnetic radiation with power P given by the equation

$$P = e\sigma AT^4$$

where e is a constant of the surface and σ is a constant given by 5.67×10^{-8} W m^{-2} K^{-4} .

(a) Show that e has no units.

[1]

(b) In an experiment to measure e using a spherical object of diameter d, the following data were obtained in Fig. 1.1

quantity	magnitude	uncertainty
P / W	6.0	± 0.2
d / cm	5.1	± 0.1
T / K	450	± 1

Fig. 1.1

(i) Explain clearly how systematic error can be reduced in the measurement of d using a pair of vernier calipers.

……

…… [1]

(ii) Determine the value of e and its actual uncertainty to the appropriate number of significant figures.

e = ... ± ..... [4]

2 **(a)** Define *gravitational potential* at a point.

…….……..

…….……..

…….…….. [1]

(b) The Earth may be assumed to be an isolated uniform sphere with its mass M concentrated at its centre. A satellite of mass m orbits the Earth in a circular path of radius R.

For the satellite in its orbit, show that its kinetic energy E_k is given by

$$E_k = \frac{GMm}{2R}.$$

[2]

(c) The Earth has radius $6.4 \times 10^6 \text{ m}$ and mass $6.0 \times 10^{24} \text{ kg}$.

A satellite has mass 850 kg and orbital radius $7.2 \times 10^6 \text{ m}$.

(i) Determine the speed of the satellite.

speed = …… m s^{-1} [2]

(ii) Determine quantitatively whether the satellite could be in a geostationary orbit.

……

…… [2]

(d) The satellite in **(c)** gradually loses energy due to small resistive forces.

Suggest why many such satellites eventually ‘burn up’ in the Earth’s atmosphere.

……

……

……

……

……

…… [3]

3 **(a)** State what is meant by the *internal energy* of a system.

……..

……..

…….…….. [1]

(b) A sample of an ideal gas at thermodynamic temperature T has internal energy U.

The gas is compressed so that its temperature increases to $3T$. During this compression, work W is done on the gas.

The gas is then cooled at constant volume so that its temperature decreases to $2T$.

Complete Table 3.1 to show, in terms of some or all of W, T and U, the work done on the gas, the thermal energy supplied to the gas and the increase in internal energy of the gas for each of the two processes.

Table 3.1

	work done on gas	thermal energy supplied to gas	increase in internal energy of gas
compression	$+W$		
cooling			

[2]

(c) A block of aluminium has a volume of 3.61×10^{-3} m^3 at a temperature of 0 °C.

Aluminium has a density of 2.70×10^{3} kg m^{-3} at 0 °C.
It has a density of 2.62×10^{3} kg m^{-3} at 500 °C.

The block is heated so that its temperature increases from 0 °C to 500 °C at an atmospheric pressure of 1.01×10^{5} Pa.

(i) Calculate the volume of the block at a temperature of 500 °C.

volume = ... m^3 [2]

(ii) Use the information in **(c)(i)** to determine the magnitude of the work done on the block when its temperature is raised from 0 °C to 500 °C.

work done = ... J [2]

(iii) Explain whether the work done on the block is positive or negative.

.....

.....

..... [2]

4 **(a)** Coherent light is incident normally on a double slit, as shown in Fig. 4.1.

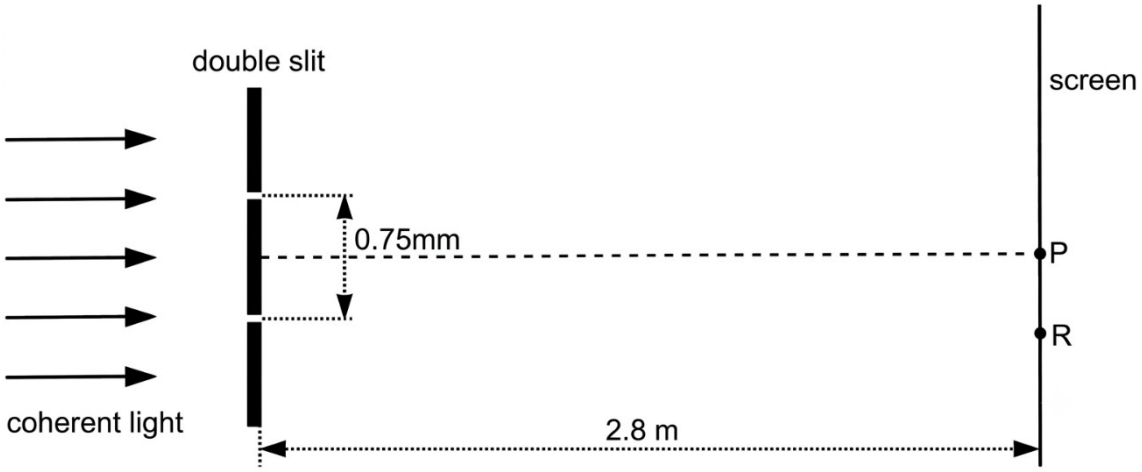


Fig. 4.1 (not to scale)

The separation of the slits in the double-slit arrangement is 0.75 mm.

A screen is placed parallel to, and at a distance of 2.8 m from, the double slit.

P is a point on the screen that is equidistant from the two slits.

Fig. 4.2 shows the variation with distance from P of the intensity I of the light on the screen.

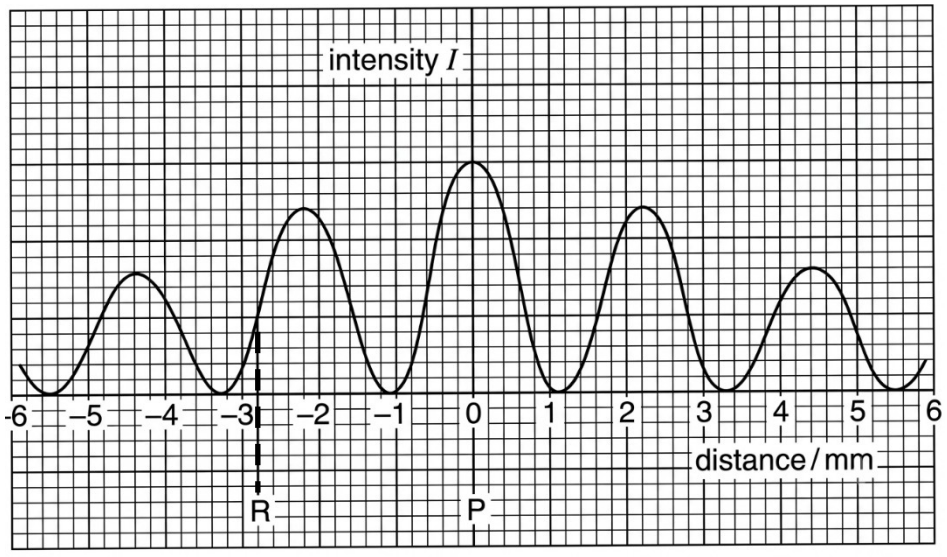


Fig. 4.2

(i) Calculate the wavelength of the coherent light.

wavelength = …… m [2]

(ii) Point R is another point on the screen. Its position is indicated on Fig. 4.2.

Determine the phase angle between the waves from the double slit when the waves meet at point R.

phase angle = …… rad [2]

(iii) Suggest why the maxima on Fig. 4.2 are not all of the same intensity.

……

……

……

…….. [2]

(b) The yellow light emitted by a sodium lamp consists of light of two wavelengths, 589.0 nm and 589.6 nm.

This light is incident normally on a diffraction grating. The spacing of the lines on the grating is 1.60×10^{-6} m.

(i) Determine the total number of directions of diffracted light in which the two wavelengths will be visible separately.

number = …… [3]

(ii) The separation of the two wavelengths in any order of the diffracted light is small.

State and explain in which order the lines have their greatest separation.

……

……

……

…… [2]

5 **(a)** By reference to heating effect, explain what is meant by the *root-mean-square* (r.m.s.) value of an alternating current.

…..

…..

….. [1]

(b) A varying current I passes through a length of wire X of resistance *R* as shown in the circuit in Fig. 5.1.

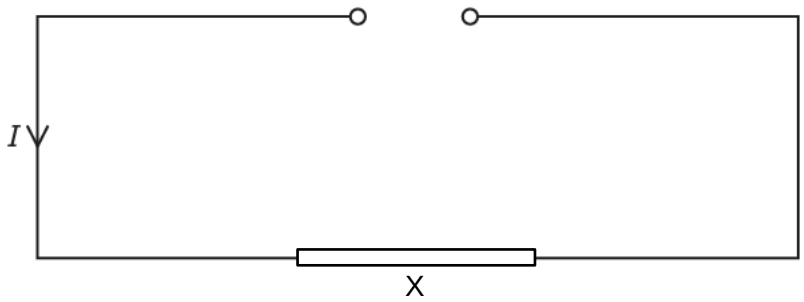


Fig. 5.1

Fig. 5.2 shows the variation with time *t* of I.

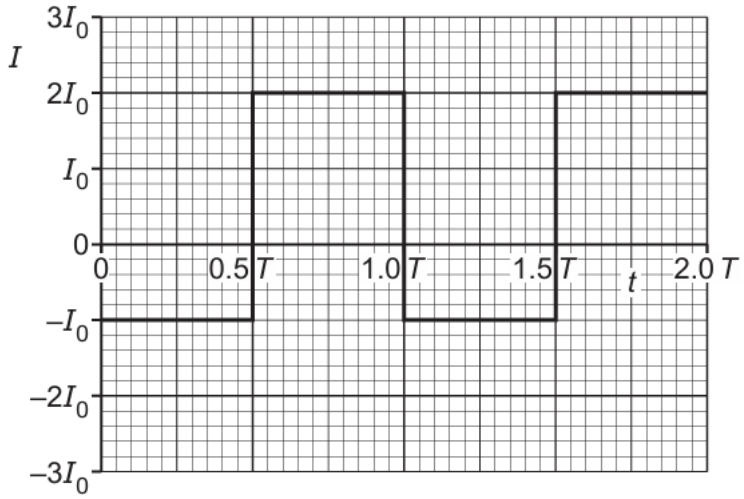


Fig. 5.2

The current has magnitude $2I_0$ when it is in the positive direction and I_0 when it is in the negative direction. The period of the variation of the current is *T*.

(i) Determine an expression, in terms of I_0, for the root-mean-square (r.m.s.) current $I_{\text{r.m.s.}}$ in wire X.

$I_{\text{r.m.s.}}$ = ... [2]

(ii) Determine an expression, in terms of I_0 and R, for the mean power $\langle P \rangle$ in wire X.

$\langle P \rangle$ = ... [1]

(iii) On Fig. 5.3, sketch the variation of the power P with t between $t = 0$ and $t = 2.0T$.

Include appropriate scale for the power axis.

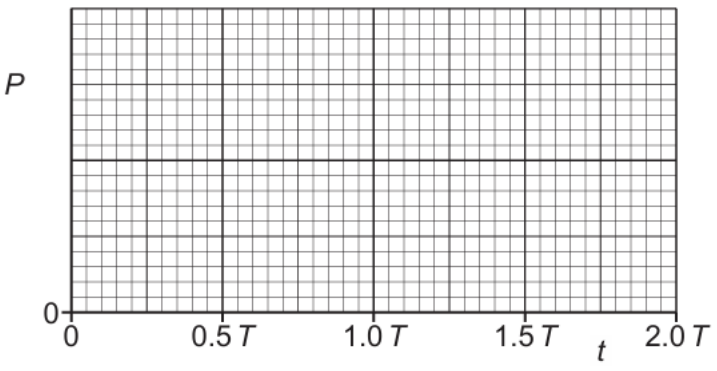


Fig. 5.3

[3]

(c) A length of wire Y, made from a material different from wire X, is connected in series with wire X, as shown in Fig. 5.4.

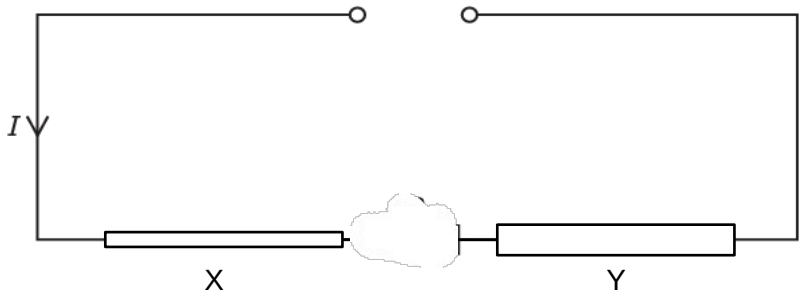


Fig. 5.4

Wire X has half the radius of wire Y.

The drift velocity of the electrons in wire X is three times the drift velocity of the electrons in wire Y.

Determine the ratio

$$\frac{\text{number density of charge carriers in Y}}{\text{number density of charge carriers in X}}.$$

Show your working.

ratio = ... [3]

6 **(a)** State what is meant by a *photon*.

……..

…….. [1]

(b) State what is meant by *threshold frequency* as applied to the photoelectric effect.

……..

……..

…….. [1]

(c) Discuss how the existence of a threshold frequency in the photoelectric effect provides evidence for the particulate nature of electromagnetic radiation.

.....

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.....

.....

..... [2]

(d) Electron diffraction may be observed using a crystal if the de Broglie wavelength of the electron is appropriate.

Calculate

(i) the de Broglie wavelength of an electron travelling with speed 5.94×10^{6} m s^{-1},

de Broglie wavelength = ... m [2]

(ii) the potential difference through which the electron must be accelerated from rest.

potential difference = ... V [2]

(iii) The separation between atoms is on the order of magnitude of 10^{-10} m.

Explain whether electrons having this speed of 5.94×10^{6} m s^{-1} will demonstrate a clear electron diffraction pattern.

.....

.....

.....

.....

.....

..... [2]

7 Read the passage below and answer the questions that follow.

In the petrochemical and natural gas industries, the ability to control the flow of high-pressure fluids is critical. Engineers use Scotch Yoke Actuators to operate large "quarter-turn" ball or butterfly valves. Unlike standard piston drives, the Scotch Yoke's unique geometry provides a high torque at the start and end of its stroke, exactly when it is needed to break a valve seal or seat it firmly.

However, mechanical vibrations and pressure surges can cause "hunting" (unstable oscillations) in these valves. To prevent mechanical failure, engineers incorporate viscous dampers and springs into the actuator assembly.

The following investigation is based on a performance test of a Scotch Yoke Actuator to ensure its damping and efficiency meet safety standards.

This performance test of a Scotch Yoke Actuator uses a Scotch Yoke Mechanism coupled to a mass, spring and damper. The setup is shown in Fig. 7.1. This is a specialised reciprocating motion mechanism used to convert the uniform circular motion of a disc into pure linear simple harmonic motion (s.h.m.).

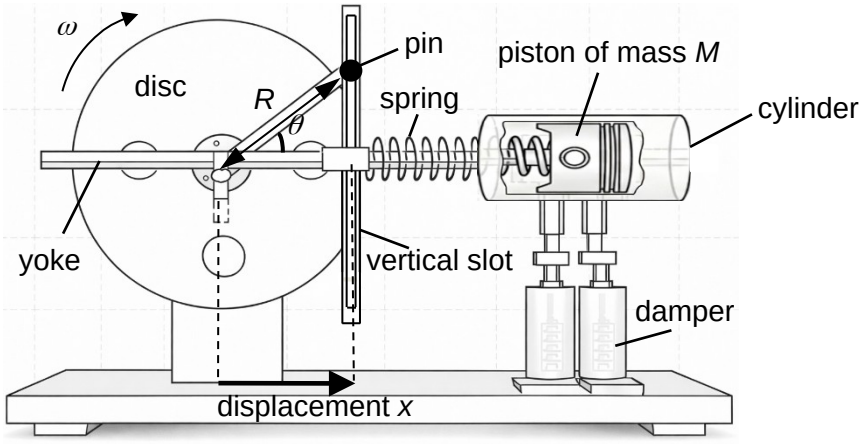


Fig. 7.1

A pin is fixed at a radial distance R from the centre of a disc rotating with a constant angular velocity ω. This pin slides within a vertical slot of a yoke assembly. As the disc rotates, the vertical slot allows the pin to move freely in the y-direction, while the yoke (the horizontal bar structure) itself is constrained to move only horizontally. Consequently, the horizontal displacement x of the yoke at any time t is exactly equal to the horizontal component of the pin's circular path.

In this setup, the yoke is attached to a piston of mass M of 2.50 kg. The piston is connected to a fixed support by a spring of force constant of 440 N m^{-1}. To simulate real-world conditions, the piston moves within a cylinder containing a viscous fluid, providing a resistive (damping) force F given by

$$F = -bv$$

where v is the velocity of the piston and b is the damping constant.

The rotating disc is driven by a motor that maintains a constant angular velocity despite energy losses in the system. As the piston oscillates, energy is continuously transferred between the rotational motion of the disc, the translational motion of the piston, and the

elastic potential energy of the spring. Some of this energy is dissipated as thermal energy due to the resistive force.

During a "Fail-Safe" test, the drive motor is disconnected to see how quickly the system returns to equilibrium. The amplitude A decays over time t according to the expression

$$A = A_0 e^{-\frac{bt}{2M}}$$

An engineer records the decay data to determine the fluid viscosity (damping constant b). The recorded data are shown in Table 7.1.

Table 7.1

t / s	0	2.0	4.0	6.0	8.0
A / m	0.120	0.066	0.036	0.020	0.011

The corresponding graph of ln(A) against t is shown in Fig. 7.2.

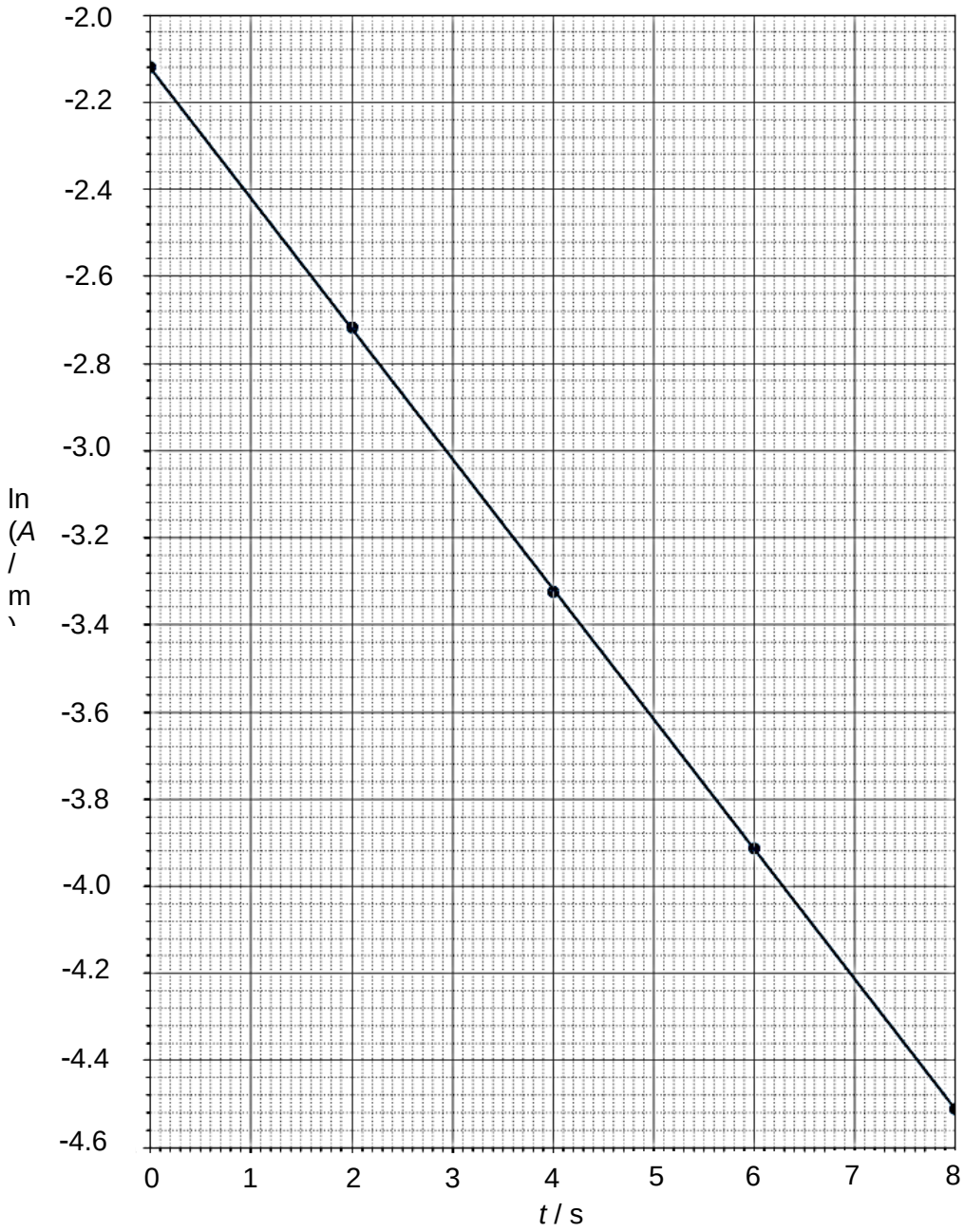


Fig. 7.2

During normal operation, the system may be subjected to periodic pressure pulses. The motor is re-engaged to drive the disc at a variable frequency f.

When the frequency of these pulses closely matches the natural frequency of the actuator, resonance occurs. Fig 7.3 shows the resonant response (amplitude of driven oscillation x_0 vs driver frequency f) for the Scotch Yoke system.

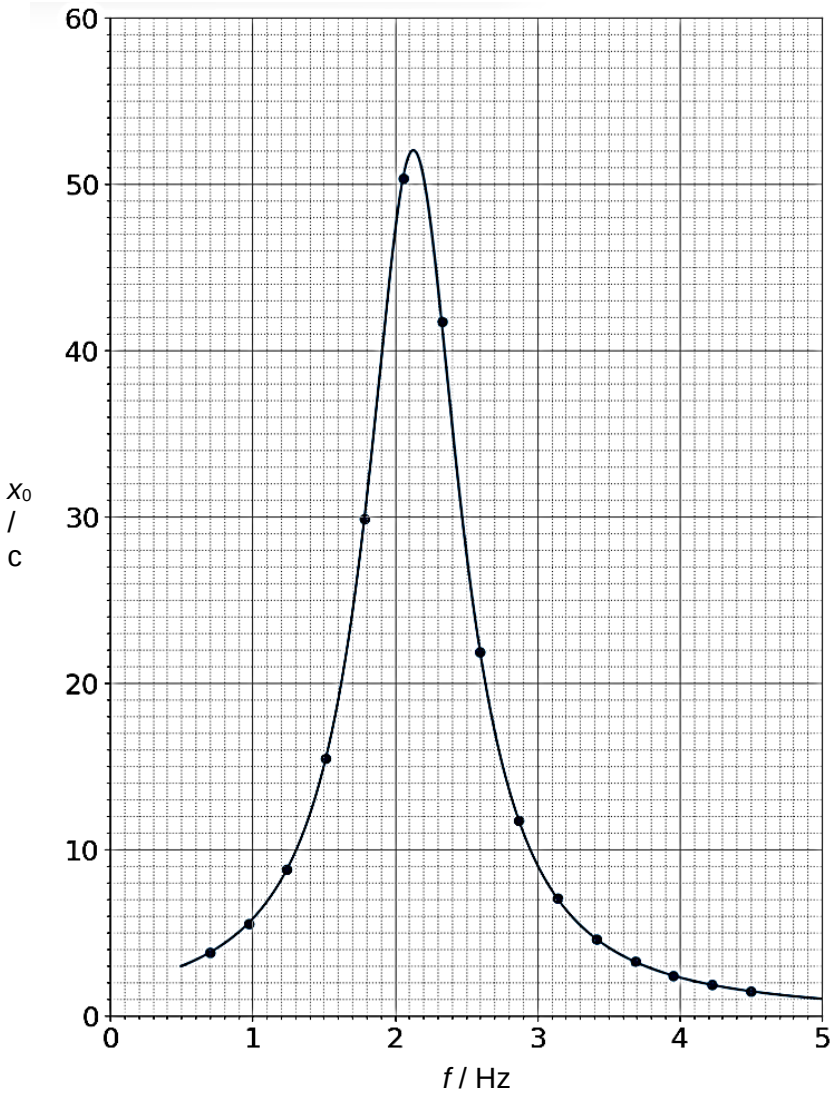


Fig. 7.3

To maintain this steady-state amplitude, the motor must supply energy at the same rate it is dissipated by the fluid. The average power dissipated is given by the expression

$$P_{\text{ave}} = \frac{1}{2} b\omega^2 x_0^2$$

In remote pipeline locations, power is often supplied by solar-charged batteries. The actuator's electric motor operates at a voltage of 24 V and draws a current of 1.8 A to maintain the resonant amplitude.

Finally, engineers often need to determine what percentage of this electrical energy is successfully converted into mechanical work against the damping fluid, versus what is lost to friction and heat in the yoke slider.

(a) (i) By considering the geometry of the rotating pin, show that the horizontal displacement of the yoke slider is given by

$$x = R\cos(\omega t).$$

[1]

(ii) State what is meant by *simple harmonic motion*.

.....

.....

..... [2]

(iii) Use **(a)(i)** to show that the yoke's motion satisfies the condition for simple harmonic motion.

[2]

(b) When the drive motor is disconnected during the "Fail-Safe" test, the system undergoes damped free oscillations.

(i) Show that a graph of ln(A) against t is a straight line.

[1]

(ii) Identify the physical significance of the gradient.

……

…… [1]

(iii) Determine the damping constant *b* for the system

b = ... kg s^{-1} [2]

(iv) Explain how the graph in Fig. 7.2 will change if the viscosity of the fluid is increased.

.....

.....

.....

..... [2]

(c) (i) The motor is re-engaged to drive the disc at a variable frequency *f*.

Use Fig. 7.3 to identify the resonant frequency and show that the spring constant is 440 N m^{-1}.

[2]

(ii) Calculate the maximum acceleration of the piston when it is operating at the resonance.

acceleration = ... m s^{-2} [2]

(d) (i) Calculate the average power dissipated by the damping force when the system is at resonance.

P_{ave} = ... W [2]

(ii) Calculate the overall efficiency of the energy transfer from the electrical supply to the mechanical oscillations if the actuator is operated in locations powered by solar-charged batteries.

efficiency = ... % [2]

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